YAOYU ZHU

Southeast University, Nanjing, China ◇ yaoyuzhu@seu.edu.cn ◇ Google Scholar

EDUCATION

Southeast University (SEU) Aug 2023 - Jun 2027 (expected)

Bachelor of Information Engineering (Chien-Shiung Wu Honor College)

GPA: 3.93/4.00 (**Rank 1/201**)

University of Cambridge Jul 2024 - Aug 2024

Visiting Student.

PUBLICATION AND PATENT

Yaoyu Zhu, Xingyu Zhou, Chenghai Huang, Le Liang, & Shi Jin. Fully Decentralized Massive MIMO Detection with Gradient-Based MCMC. In *Proc. IEEE 37th Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, pp. 1–6, 2026.

Yaoyu Zhu, Xingyu Zhou, Chenghai Huang, Le Liang, & Shi Jin. Gradient-Based MCMC for Fully Decentralized Massive MIMO Detection. *Physical Communication*, vol. 80, 103333, 2026.

Le Liang, **Yaoyu Zhu**, Xingyu Zhou, Chenghai Huang, & Shi Jin. A Fully Distributed Detection Method Using Markov Chain Monte Carlo with Message Passing (in Chinese). Chinese Patent, factual check phase.

AWARDS AND HONORS

National Scholarship (Top 0.5%, Ministry of Education of China) x2 (2024, 2025)

Taihu Future Technology Scholarship (2024, 5000RMB)

Zhishan Student Scholarship (2024, 2000 RMB)

Chien-Shiung Student (2024, 10/1000+ students, voted annually as highest honor of CSW College)

National Grand Prize, National English Competition for College Students (NECCS)

Meritorious Winner, Mathematical Contest in Modeling (MCM)

SKILLS

Languages: English (**IELTS: 8.0**(speaking & writing: 7.0)), Mandarin (Native), Cantonese, Teochew Language

Programming Languages: Python / MATLAB / C++ / C

Python Packages: NumPy / SciPy / sklearn / PyTorch / TensorFlow / Keras

Technical Skills: LaTeX/ Markdown

RESEARCH EXPERIENCE

Wearable Agents and Egocentric Reasoning Jul 2025 - Aug 2025

Hong Kong University of Science and Technology (HKUST) | **Research Intern** | **Advisor: Prof. Jun Zhang**

- Explored wearable-agent systems and built a chess demo for RayNeo smart glasses, implementing the TCP communication pipeline between the device and backend services.
- Curated and annotated EgoLife data to construct a new benchmark covering spatial reasoning and related egocentric reasoning capabilities.

Towards Intelligent Receivers: A Markov Chain Monte Carlo Approach Oct 2024 - Sept 2025

Southeast University | **Project Leader** | **Advisor: Prof. Le Liang**

- Derived a Gradient-Based MCMC sampling strategy utilizing **exponential family approximations**, reducing computational complexity and interconnection bandwidth while achieving near-optimal performance.
- Implemented the algorithm in Python with numpy, scikit-learn etc. and conducted extensive simulations to validate its robustness against channel noise.

SELECTED COMPETITIONS

Mathematical Contest in Modeling (MCM), Meritorious Winner Jan 2025

Unlocking the Medal Count: Predictive Models and Impact of Great Coaches on the 2028 Olympics. Developed an Entropy Weight Method-based ability index with regression and GM(1,1) models for medal forecasting, and quantified coaching impact using regression and random forests.